BIRLA INSTITUTE OF TECHNOLOGY

NEP-2020 CURRICULUM BOOK

(Effective from Academic Session: Monsoon 2024)

B.TECH IN ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Institute Vision

To become a Globally Recognised Academic Institution in consonance with the social, economic and ecological environment, striving continuously for excellence in education, research, and technological service to the National needs.

Institute Mission

- To educate students at Under Graduate, Post Graduate, Doctoral, and Post-Doctoral levels to perform challenging engineering and managerial jobs in industry.
- To provide excellent research and development facilities to take up Ph.D. programmes and research projects.
- To develop effective teaching learning skills and state of art research potential of the faculty.
- To build national capabilities in technology, education, and research in emerging areas.
- To provide excellent technological services to satisfy the requirements of the industry and overall academic needs of society.

Department Vision

The department strives to be recognized globally for outstanding education and research, leading to excellent professionals and innovators in the field of Computer Science and Engineering, who can positively contribute to the society.

Department Mission

- To impart quality education and equip the students with strong foundation that could make them capable of handling challenges of the new century.
- To maintain state of the art research facilities and facilitate interaction with world's leading universities, industries and research organization for constant improvement in the quality of education and research.

Programme Educational Objectives (PEOs)

The program aims to aid students:

- To possess technical skills adoptable to complex Computer Science & Engineering problems in order to lead in their domain.
- To enhance one's academic credentials and grow to be a leading luminary.
- To develop into a job creator and flourish further the national agenda.
- To inculcate ethical responsibility towards the society and impact optimistically, the social fabric of the nation.

Programme Outcomes (POs)

Engineering Graduates will be able to:

- Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

- Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
- Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
- Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
- The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
- Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
- Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
- Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
- Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
- Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
- Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Programme Specific Outcomes (PSOs)

- PSO 1: Analyze and apply knowledge of machine learning, Artificial intelligence and data engineering to solve real world problems.
- PSO 2: Develop computational knowledge and project development skills to solve research problems related to machine learning and artificial intelligence.
- PSO 3: Inculcate best practices and principles in consideration with existent societal, cultural, environmental and financial aspects.

SEMESTER 1 - FIRST YEAR

Total Credits: 22

Course Code	Course Name	L	T	P	Credits	Category
MA24101	Mathematics - I	3	1	0	4	FS
CH24101	Chemistry	3	1	0	4	GE
EC24101	Basic Electronics and Communication Engineering	2	1	0	3	GE
ME24101	Basics of Mechanical Engineering	2	1	0	3	FS
CE24101	Environmental Sciences	2	0	0	2	FS
CH24102	Chemistry Lab	0	0	2	1	FS
EC24102	Basic Electronics and Communication Lab	0	0	2	1	GE
ME24102	Engineering Graphics	0	0	4	2	GE
PE24102	Workshop Practice	0	0	2	1	GE
MC24101-105	Choice: NCC/NSS/PT & Games/Creative Arts/Entrepreneurship	0	0	2	1	MC

SEMESTER 2 - FIRST YEAR

Total Credits: 22.5

Course Code	Course Name	L	T	P	Credits	Category
MA24103	Mathematics - II	3	1	0	4	FS

PH24101	Physics	3	1	0	4	FS
BE24101	Biological Sciences for Engineers	2	0	0	2	GE
CS24101	Programming for Problem Solving	3	1	0	4	GE
EE24101	Basics of Electrical Engineering	2	1	0	3	GE
PH24102	Physics Lab	0	0	2	1	FS
CS24102	Programming for Problem Solving Lab	0	0	2	1	GE
EE24102	Electrical Engineering Lab	0	0	2	1	GE
HS24131	Communication Skill - I	0	0	3	1.5	HSS
MC24106-110	Choice: NCC/NSS/PT & Games/Creative Arts/Entrepreneurship	0	0	2	1	MC

Grand Total for First Year: 44.5 Credits

SEMESTER 3 - SECOND YEAR

Total Credits: 24

Course Code	Course Name	L	T	P	Credits	Category
CS24201	Data Structures and Algorithms	3	0	0	3	PC
CS24205	Computer Organization and Architecture	3	0	0	3	PC
AI24101	Mathematics for AI-ML	3	0	0	3	PC
MA24205	Discrete Mathematics	3	0	0	3	PC
EC24203	Digital System Design	3	0	0	3	PC
HS24131	UHV-II: Understanding Harmony	3	0	0	3	HSS
CS24202	Data Structures and Algorithms Lab	0	0	3	1.5	PC
AI24102	Mathematics for AI-ML Lab	0	1	3	2.5	PC
EC24204	Digital System Design Lab	0	0	2	1	PC
MC24201-205	Choice: NCC/NSS/PT & Games/Creative Arts/Entrepreneurship	0	0	2	1	MC

SEMESTER 4 - SECOND YEAR

Total Credits: 25.5

Course Code	Course Name	L	T	P	Credits	Category
CS24211	Database Management System	3	0	0	3	PC
CS24213	Design and Analysis of Algorithm	3	0	0	3	PC
CS24215	Operating System	3	0	0	3	PC
AI24111	Introduction to AI	3	1	0	4	PC
OE/MOOC	Open Elective - I / MOOC - I	3	0	0	3	OE
MA24201	Numerical Methods	2	0	0	2	PC
HS24211	Indian Knowledge System	2	0	0	NC	HSS
CS24212	Database Management System Lab	0	0	3	1.5	PC
CS24216	Shell and Kernel Programming Lab	0	0	3	1.5	PC
CS24218	Advanced Programming Lab	0	1	3	2.5	PC
MA24202	Numerical Methods Lab	0	0	2	1	PC

Grand Total for Second Year: 49.5 Credits

SEMESTER 5 - THIRD YEAR

Total Credits: 26

Course Code	Course Name	L	T	P	Credits	Category
AI24301	Natural Language Processing	3	0	0	3	PC
CS24303	Data Mining Concepts and Techniques	3	0	0	3	PC
CS24305	Data Communication and Computer Networks	3	0	0	3	PC
AI24303	Machine Learning Techniques	3	0	0	3	PC
PE	Program Elective - I (PE-I)	3	0	0	3	PE
OE/MOOC	Open Elective - II / MOOC - II	3	0	0	3	OE
AI24302	Natural Language Processing Lab	0	0	3	1.5	PC
CS24306	Data Comm. and Computer Networks Lab	0	0	3	1.5	PC
AI24304	Machine Learning Techniques Lab	0	0	3	1.5	PC
PE Lab	Program Elective - I Lab	0	0	3	1.5	PE
AI24350	Project - I				2	PC

HS24133	Communication Skill - II	0	0	3	1.5	HSS

SEMESTER 6 - THIRD YEAR

Total Credits: 25

Course Code	Course Name	L	T	P	Credits	Category
AI24313	Deep Learning	3	0	0	3	PC
CS24315	Introduction to Cyber Security	3	0	0	3	PC
PE	Program Elective - II (PE-II)	3	0	0	3	PE
PE	Program Elective - III (PE-III)	3	0	0	3	PE
OE/MOOC	Open Elective - III / MOOC - III	3	0	0	3	OE
HS24204	Constitution of India	2	0	0	NC	HSS
AI24314	Deep Learning Lab	0	0	3	1.5	PC
CS24318	Embedded Systems Lab	0	0	3	1.5	PC
PE Lab	Program Elective - II Lab	0	0	3	1.5	PE
AI24350	Project - II				2	PC

Grand Total for Third Year: 51 Credits

SEMESTER 7 - FOURTH YEAR

Total Credits: 17.5

Course Code	Course Name	L	T	P	Credits	Category
PE	Program Elective - IV (PE-IV)	3	0	0	3	PE
PE	Program Elective - V (PE-V)	3	0	0	3	PE
OE/MOOC	Open Elective - IV / MOOC - IV	3	0	0	3	OE
PE Lab	Program Elective - IV Lab	0	0	3	1.5	PE
MC24400	Summer Training (Minimum 4 Weeks / 160 Hrs)				4	MC
AI24400	Project - III				3	PC

SEMESTER 8 - FOURTH YEAR

Total Credits: 19

Course Code	Course Name	L	T	P	Credits	Category
AI24411	Departmental Research & Software Development	3	0	0	3	PC
AI24415	Project - IV / Industry Internship	0	0		6	PC
AI24498	Comprehensive Viva	0	0		2	PC

Grand Total for Fourth Year: 36.5 Credits

GRAND TOTAL FOR B.TECH: 168 Credits

LIST OF PROGRAM ELECTIVES

PE	Course Code	Course Name	Credits
PE-I	AI24351	Formal Language and Automata Theory	3
	CS24353	Software Engineering	3
	CS24361	Information Retrieval	3
	CS24362	Information Retrieval Lab	1.5
PE-II	AI24361	Evolutionary Computing	3
	AI24362	Evolutionary Computing Lab	1.5
	CS24365	Image Processing	3
	CS24366	Image Processing Lab	1.5
PE-III	AI24371	Introduction to Compiler Design	3
	AI24373	Statistical Machine Learning	3
	AI24375	Pattern Recognition	3
PE-IV	CS24453	Computer Vision	3
	CS24454	Computer Vision Lab	1.5
	CS24455	Internet of Things	3

	CS24456	Internet of Things Lab	1.5
PE-V	AI24457	Distributed Systems	3
	AI24458	Distributed Systems Lab	1.5
	CS24459	Optimization Techniques	3
	CS24460	Optimization Techniques Lab	1.5
	CS24471	Cloud Computing	3
	AI24471	Speech Processing	3
	CS24475	Cryptography and Network Security	3